

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 9: L14

Question 1 (Lecture 14)

True or false: While the underrepresentation of specific groups in training data is a common contributor to data bias, addressing this imbalance alone is always sufficient to ensure equitable and unbiased performance of an AI model in real-world deployment scenarios.

- (a) True
- (b) False

Question 2 (Lecture 14)

Which of the following best describes model bias?

- a) A biased dataset that leads to skewed predictions
- b) A mismatch between training and deployment settings
- c) A model that makes incorrect or overly complex assumptions, either oversimplifying or overfitting the data
- d) A model trained using synthetic data

Question 3 (Lecture 14)

True or false: Model bias can occur even if the training data is perfectly balanced.

- (a) True
- (b) False

Question 4 (Lecture 14)

Which of the following practices most effectively mitigates model bias at its source?

- a) Employing rigorous data standardization techniques to ensure all input features are scaled uniformly across the dataset
- b) Maximizing the complexity of the chosen model architecture to allow for the

learning of highly intricate and nuanced patterns within the training data

- c) Consistently enriching the training dataset with new, contextually diverse, and representative examples, followed by iterative model retraining
- d) Implementing post-hoc fairness algorithms to adjust the model's outputs only after initial predictions are generated, without altering the training process.

Question 5 (Lecture 14)

Which of the following is a key cause of deployment bias?

- a) Using outdated training data
- b) Selecting an overly simple model
- c) Applying the model in a context very different from its training conditions
- d) Ignoring missing values during data preprocessing

Question 6 (Optional – For Exploration)

An engineering firm is developing a new Large Language Model (LLM) application for use in confidential industrial design and simulation workflows. This LLM will process sensitive proprietary data and potentially intellectual property during its operation and refinement. The firm is acutely aware of the data privacy challenges inherent in LLMs.

Consider the following common privacy-enhancing techniques employed in LLM development and deployment:

- Data Anonymization
- Differential Privacy
- Access Controls and Filters
- Federated Learning / On-device processing

Select any two of these techniques. For each chosen technique, provide a detailed discussion of the key engineering **trade-offs** that must be evaluated during its design and implementation. Your analysis should encompass considerations such as:

--The impact on the LLM's performance (e.g., accuracy, model utility, inference speed).

--The user experience for end-users interacting with the LLM (e.g., ease of use, perceived privacy, functionality limitations).

--The system complexity introduced for development, integration, and maintenance.

--The guaranteed level of privacy assurance versus the feasibility and implementation cost.